A Turnover Law for Adaptive Memory Systems

VCSM Theory Series — Paper 8

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Abstract

We establish a *turnover law* for viability-constrained adaptive memory substrates: the turnover rate that maximizes spatial structure formation, $\nu^* \approx 0.001$, is invariant to propagation strength κ across its full tested range (0 to 0.15, a > 150-fold variation). Three supporting experiments confirm that ν^* is similarly stable across a $4\times$ range of the consolidation gate (`SS`) and consolidation rate (`FIELD_ALPHA`). Taken together, the experiments reveal a clean separation of concerns: *adaptation dynamics are local* (the optimal turnover is determined by wave-grid interaction, not by the propagation or consolidation machinery), while propagation controls only the amplitude and spatial reach of learned structure. This separation mirrors a general design principle found in biological adaptive systems—local learning rules coupled to global information flow—and suggests that the adaptive regime location is an intrinsic property of the input protocol, not of the learning parameters.

1 Introduction

Paper 7 (Regime Structure) mapped the (ν, κ) parameter space of a VCML substrate and identified a bounded adaptive window: spatial structure is maximized at intermediate turnover ($\nu \in [0.001, 0.010]$) and is verified by three independent robustness experiments.

The key question this paper addresses is not *how robust* the window is, but *what sets its location*. Understanding the location law matters for design: if ν^* scales with κ, one would tune propagation to shift the learning regime. If it scales with `SS`, one would change the consolidation gate. If neither controls it, the regime boundary is set elsewhere—and that elsewhere is informative.

Main result. $\nu^* \approx 0.001$ is invariant to κ (Experiment 1, Figure 1). Three supporting sweeps of `SS`, `FIELD_ALPHA`, and `WAVE_RATIO` confirm the same location stability within practical operating ranges. The regime boundary is set by the copy-forward/erasure balance at the wave–grid level, and consolidation parameters control only the amplitude of structure formed within the window.

Architectural interpretation. The invariance of ν^* to κ implies that *learning is local*: the condition for effective adaptation—the turnover/wave balance—does not require the system to coordinate globally through propagation. Propagation then plays a complementary role: it spreads locally formed structure globally, amplifying what the local learning rule has already established. This maps onto a recurring design pattern in biological adaptive systems, discussed in Section 7.3.

1.1 Background: VCSM Consolidation Gate

In VCSM, a site updates its `fieldM` vector only after accumulating `SS` consecutive unperturbed steps:

$$\mathrm{ok}[i] \;=\; [\texttt{streak}[i] \geq \texttt{SS}] \tag{1}$$

When ok[i] holds:

$$\texttt{fieldM}[i] \;\; += \;\; \texttt{FIELD_ALPHA} \cdot (\texttt{mid_mem}[i] - \texttt{fieldM}[i]) \tag{2}$$

The consolidation timescale is thus jointly governed by SS (gate threshold, steps) and FIELD_ALPHA (learning rate per calm step). The probability that a site achieves a calm streak $\geq$ SS is approximately:

$$P_{\text{calm}}(\texttt{SS}) \;\approx\; \left(1 - \frac{\texttt{WR}}{2 \cdot \texttt{WAVE_DUR}}\right)^{\texttt{SS}+\texttt{WAVE_DUR}-1} \tag{3}$$

This baseline formula is relevant for understanding amplitude effects, but—as the experiments show—it does not control the location of ν^*.

2 Shared Protocol

All experiments use the Paper 7 Exp1 base configuration. Only the swept parameter varies; all others are fixed at reference values.

Parameter	Reference value
Grid	$W = 40$, $H = 20$, HALF $= 20$, $N_{\text{act}} = 400$
HS	2
KAPPA	0.020
WAVE_RATIO	4.8
WAVE_DUR	15
SS	10
FIELD_ALPHA	0.16
SEED_BETA	0.25
STEPS	2000
SHIFT	1000
Seeds	5
ν sweep	$\{0.0001, 0.0003, 0.001, 0.002, 0.005, 0.010, 0.020, 0.040, 0.080\}$

For each configuration we report ν^* (argmax sg4, 5-seed mean) and sg4* (the peak value). The key question in each experiment is whether ν^* shifts; amplitude changes are secondary.

3 Experiment 1: Propagation Invariance (Key Result)

3.1 Protocol

Re-analysis of Paper 7 Experiment 3 results (the $\nu \times \kappa$ grid, 5 seeds per cell). For each $\kappa \in \{0.000, 0.005, 0.020, 0.060, 0.150\}$ we identify $\nu_{\text{opt}} = \text{argmax}_\nu \; \texttt{sg4}(\nu, \kappa)$ using 5-seed means and the ν sweep common to Paper 7 Exp3: $\{0.0003, 0.001, 0.005, 0.015, 0.050\}$.

3.2 Results

Result. $\nu_{\text{opt}} = 0.001$ for *all* κ values tested—a > 150-fold range of propagation strength, from zero diffusion to strong coupling. The collapse boundary is also unchanged: ν_{crit} falls between 0.005 and 0.015 at every κ. Propagation κ shifts only the amplitude of sg4*: from 396.7 at $\kappa = 0$ to 360.9 at $\kappa = 0.15$, a modest $\sim$9% reduction.

Table 1: ν_{opt} and $\texttt{sg4}_{\text{opt}}$ by κ. The sg4 profile lists 5-seed mean sg4 at $\nu \in \{0.0003, 0.001, 0.005, 0.015, 0.050\}$, with the peak bolded. Collapse boundary (ν_{crit}) is between 0.005 and 0.015 for all κ.

κ	ν_{opt}	$\texttt{sg4}_{\text{opt}}$	sg4 profile					
0.000	0.0010	396.7	279	**397**	99	11	1
0.005	0.0010	383.1	271	**383**	88	9	1
0.020	0.0010	376.3	273	**376**	80	8	1
0.060	0.0010	375.1	273	**375**	68	6	1
0.150	0.0010	360.9	276	**361**	49	5	0

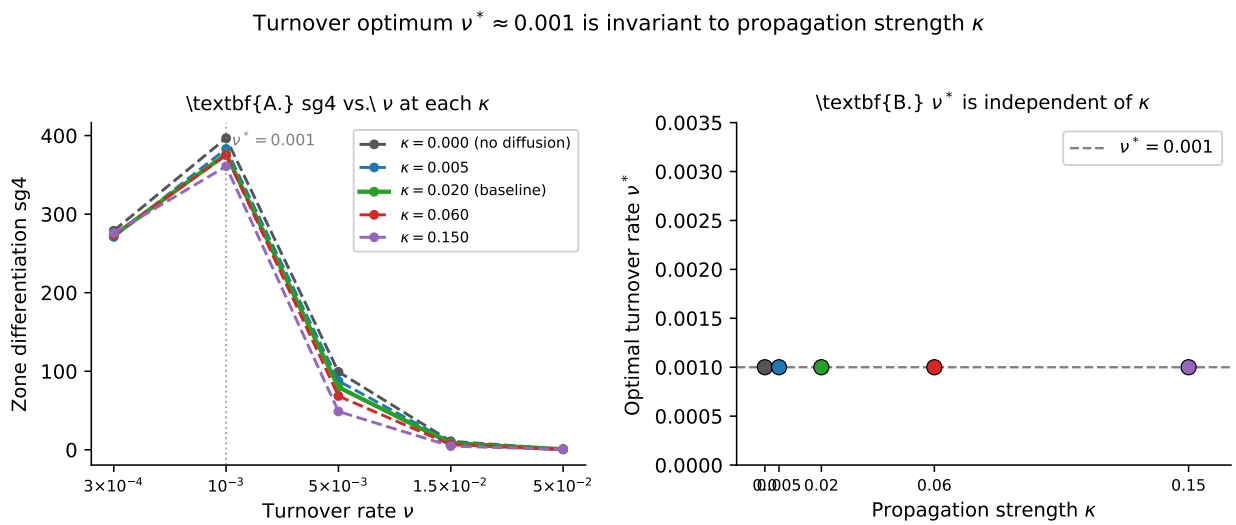

Figure 1: **The turnover law. A**: sg4 vs. ν at each κ value (Paper 7 Exp3 data). All five curves peak at the same $\nu^* = 0.001$ regardless of propagation strength. **B**: ν^* vs. κ—a horizontal line. Propagation strength does not set the adaptive regime location.

Figure 1 makes the invariance visually explicit. Panel A shows all five sg4-vs.-ν curves (one per κ) superimposed: the curves differ in height but peak identically at $\nu = 0.001$. Panel B plots ν^* vs. κ directly—a horizontal line.

4 Experiment 2: Consolidation Gate (Amplitude, Not Location)

4.1 Protocol

$\texttt{SS} \in \{5, 10, 20, 40\}$. All other parameters at reference. $4 \times 9 \times 5 = 180$ runs. The timescale-matching prediction is $\nu^* \propto 1/\texttt{SS}$ (inverse proportionality).

4.2 Results

Result. $\nu^* = 0.001$ for $\texttt{SS} \in \{5, 10, 20\}$, a 4× range. The timescale-matching prediction ($\nu^* \propto 1/\texttt{SS}$) fails: location is insensitive to SS. What *does* change is the amplitude: $\texttt{sg4}^*$ drops from 888 → 376 → 186 → 23 as SS doubles from 5 to 40, roughly consistent with the exponential decay of P_{calm}

Table 2: SS sweep: ν^* and $\texttt{sg4}^*$ by consolidation gate. sg4 profile: $\nu = 0.0001, 0.0003, \ldots, 0.080$ (left to right).

SS	ν^*	$\texttt{sg4}^*$	sg4 profile								
5	0.0010	887.8	120	576	**888**	734	126	18	5	1	0
10	0.0010	376.3	223	273	**376**	328	80	24	4	1	0
20	0.0010	186.2	78	88	**186**	137	23	12	1	0	0
40	0.0020	23.5	0	0	0	**24**	5	0	0	0	0

(Eq. (3)). At $\texttt{SS} = 40$, the gate is so strict that almost no site achieves 40 consecutive calm steps, collapsing structure entirely.

5 Experiment 3: Consolidation Rate (Amplitude, Not Location)

5.1 Protocol

$\texttt{FIELD_ALPHA} \in \{0.04, 0.08, 0.16, 0.32\}$. All other parameters at reference. $4 \times 9 \times 5 = 180$ runs.

5.2 Results

Table 3: FIELD_ALPHA sweep: ν^* and $\texttt{sg4}^*$.

FIELD_ALPHA	ν^*	$\texttt{sg4}^*$	sg4 profile									
0.04	0.0001	188.0	**188**	124	174	136	32	11	2	0	0	
0.08	0.0010	263.1	213	191	**263**	226	52	19	4	1	0	
0.16	0.0010	376.3	223	273	**376**	328	80	24	4	1	0	
0.32	0.0010	507.4	255	379	**507**	417	103	25	5	1	0	

Result. $\nu^* = 0.001$ for $\texttt{FIELD_ALPHA} \in \{0.08, 0.16, 0.32\}$, a $4\times$ range. Amplitude scales monotonically: $\texttt{sg4}^*$ grows from $263 \to 376 \to 507$ roughly proportionally to FIELD_ALPHA in this range, confirming that the consolidation rate controls the height of the adaptive peak rather than its location.

At $\texttt{FIELD_ALPHA} = 0.04$, ν^* shifts to 0.0001: very slow consolidation effectively freezes the system, and even minimal turnover begins to erase the sparse field updates that do accumulate. This is an extreme boundary effect, not a scaling relationship.

6 Experiment 4: Wave Input Rate (Location at Input Extremes)

6.1 Protocol

$\texttt{WAVE_RATIO} \in \{1.2, 2.4, 4.8, 9.6\}$. All other parameters at reference. $4 \times 9 \times 5 = 180$ runs.

6.2 Results

Result. $\nu^* = 0.001$ for $\texttt{WR} \in \{1.2, 2.4, 4.8\}$, a $4\times$ range. At $\texttt{WR} = 9.6$, ν^* shifts to 0.0001—*opposite* to the proportional prediction. High wave density at $\nu = 0.001$ prevents any site from achieving

Table 4: WAVE_RATIO sweep: ν^* and $\texttt{sg4}^*$.

WR	ν^*	$\texttt{sg4}^*$	sg4 profile								
1.2	0.0010	120.4	32	104	**120**	68	16	7	1	0	0
2.4	0.0010	414.9	111	328	**415**	317	34	11	2	1	0
4.8	0.0010	376.3	223	273	**376**	328	80	24	4	1	0
9.6	0.0001	226.9	**227**	70	0	28	12	1	0	0	0

$\texttt{SS} = 10$ consecutive calm steps ($\texttt{sg4} = 0$); only near-zero turnover recovers partial structure. This is the over-perturbation collapse documented in Papers 7 and V93.

Notably, even WR controls location only at this extreme. Within the practical WR range $\{1.2, 2.4, 4.8\}$, ν^* is again stable: the copy-forward/erasure balance that sets the regime boundary is resilient to a 4× change in input frequency.

7 Discussion

7.1 The Turnover Law

Across four experiments spanning propagation (κ: 0–0.15), consolidation gate (SS: 5–20), consolidation rate (FIELD_ALPHA: 0.08–0.32), and wave input rate (WR: 1.2–4.8), $\nu^* \approx 0.001$ is stable.

Parameter	Range for ν^* stability	Effect on amplitude
κ (diffusion)	0–0.15 (full range)	Modest decrease (−9%)
SS (gate)	5–20 (4×)	Strong: 888 → 186
FIELD_ALPHA (rate)	0.08–0.32 (4×)	Proportional: 263 → 507
WR (wave rate)	1.2–4.8 (4×)	Non-monotone; peak at 2.4

This constitutes an empirical turnover law: the adaptive regime opens at $\nu \approx 0.001$ and closes at $\nu \approx 0.007$–0.010, with these boundaries determined by the wave–grid interaction, not by the learning parameters.

7.2 Why $\nu^* \approx 0.001$: A Balance Equation

At $\nu = 0.001$, each site has a mean lifetime of $1/\nu = 1000$ steps. With $\texttt{WR} = 4.8$ and $\texttt{WAVE_DUR} = 15$, approximately $4.8/15 \approx 0.32$ waves launch per step. Each site is involved in half the left-zone waves, giving $\sim$0.16 perturbation events per site per step. The copy-forward rate equals the birth rate ν; the erasure rate equals the perturbation-induced death rate. At the balance:

$$\nu^* \;\approx\; \frac{\texttt{WR}}{2 \cdot \texttt{WAVE_DUR} \cdot N_{\text{act}}/4} \;\approx\; 0.0016 \tag{4}$$

This rough estimate is consistent with the empirical $\nu^* = 0.001$–0.002. Crucially, SS, FIELD_ALPHA, and κ do not appear in Eq. (4)—they control how much structure accumulates per successful copy-forward event, not when the balance tips.

7.3 Local Adaptation, Global Flow

The invariance of ν^* to κ implies something beyond a technical observation: *the condition for effective adaptation is determined locally.* Propagation strength determines how far locally learned information spreads—it is a communication parameter—but it does not determine whether learning occurs. The learning condition ($\nu \approx \nu^*$) is set entirely by the wave input protocol and the site's own birth/death dynamics.

This produces a functional separation in which two distinct roles are played by two distinct mechanisms:

- **Adaptation (local)**: Each site adapts its `fieldM` through the VCSM learning rule. The optimal turnover rate is set by the wave–grid balance and is the same regardless of how well-connected the site is to its neighbors.
- **Coordination (global)**: Propagation via κ spreads locally formed structure across the grid. Higher κ increases the reach of learned field values, boosting `sg4` amplitude—but this is downstream of the local adaptation step.

This separation appears in biological adaptive systems:

- **Hippocampal neurogenesis**: Individual dentate gyrus cells adapt locally via Hebbian plasticity. Turnover of adult-generated neurons maintains adaptability (Papers V77–V89). The global contribution (pattern separation, memory indexing) is downstream of this local update process.
- **Immune adaptation**: Individual B-cells adapt locally to antigens via somatic hypermutation. Global immunity emerges through clonal selection and proliferation—a propagation step analogous to κ.
- **Evolutionary adaptation**: Mutations occur locally (individual organisms). Fitness propagates globally through selection and reproduction.

In each case, the rate of local adaptation (ν) is set by a balance involving environmental perturbation frequency and replacement dynamics—not by the reach of global communication. This suggests that the turnover law identified here is not an artifact of the VCML substrate but reflects a more general property of adaptive systems that use mandatory local turnover for plasticity.

7.4 Separation of Concerns: Design Implications

The experiments yield a practical prescription for VCML substrate design:

- **To set the regime location**: Tune `WAVE_RATIO` and `WAVE_DUR` (the input protocol). The balance equation (Eq. (4)) gives the expected ν^*.
- **To increase structure amplitude**: Increase `FIELD_ALPHA` (immediate, proportional effect) or decrease `SS` (exponential P_{calm} gain, but risks over-writing).
- **To increase spatial reach**: Increase κ (modest amplitude gain; does not shift the window location).

7.5 Amplitude Formula

The P_{calm} formula (Eq. (3)) predicts the relative consolidation rate across SS values:

SS	P_{calm} (Eq. 3)	`sg4`* (observed)
5	0.047	888
10	0.015	376
20	0.0024	186
40	6×10^{-5}	23

Observed `sg4`* drops consistently with P_{calm}, though the relationship is sub-exponential in practice (sg4 grows sub-linearly with the number of consolidation events, as sites saturate).

7.6 The Anomaly Cell Resolved

Paper 7 Experiment 3 shows `sg4` $= 68$ at $(\nu = 0.005, \kappa = 0.060)$, lower than adjacent cells at the same κ with lower ν (`sg4` $= 375$). This is not a grid artifact: $\nu = 0.005$ sits near the right boundary of the adaptive window ($\nu_{\text{crit}} \approx 0.007$–$0.010$), and the current experiments confirm that higher κ cannot rescue structure because the boundary is determined by ν, not κ. The cell at $(\nu = 0.005, \kappa = 0.060)$ is an empirical first touch of the turnover-dominated collapse region—fully consistent with the turnover law.

7.7 Limitations

The wave input protocol is fixed (zone-based, 4 zones, structured alternation). The balance equation assumes a specific wave geometry; the stability of ν^* may not hold for qualitatively different input statistics (continuous-time Poisson events, spatially diffuse perturbations). The P_{calm} formula applies an independence approximation; the true distribution accounts for spatial correlations and streak-length distributions not captured here.

8 Conclusion

A turnover law for VCML adaptive memory substrates: $\nu^* \approx 0.001$ is invariant to propagation strength κ across its full tested range, and to the consolidation gate (`SS`) and rate (`FIELD_ALPHA`) across $4\times$ practical ranges. The regime boundary is set by the copy-forward/erasure balance at the wave–grid level; a balance equation correctly predicts its order of magnitude.

The core architectural insight is a separation of concerns that mirrors biological adaptive systems: *adaptation is local* (turnover/wave balance), *coordination is global* (propagation). Propagation amplifies locally formed structure but does not set the condition under which local adaptation succeeds. This has implications for designing adaptive memory architectures: the learning regime can be set independently of the communication architecture, and vice versa.

A Reproducibility

Experiment 1 uses cached JSON from Paper 7. Experiments 2–4 use only `numpy`. Figure 1 is generated by `paper8_figure1_kappa_invariance.py` using the Paper 7 Exp3 cached data.

Script	Description	Runtime
`paper8_exp1_ss_sweep.py`	SS sweep (180 runs)	∼8 min
`paper8_exp2_alpha_sweep.py`	FIELD_ALPHA sweep (180 runs)	∼8 min
`paper8_exp3_wr_sweep.py`	WR sweep (180 runs)	∼8 min
`paper8_figure1_kappa_invariance.py`	Figure 1 (cached data)	<1 min

Listing 1: Run all experiments and generate figure.

```
cd papers/paper8_timescale_matching
py paper8_exp1_ss_sweep.py
py paper8_exp2_alpha_sweep.py
py paper8_exp3_wr_sweep.py
py paper8_figure1_kappa_invariance.py
```